

HW3

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Hilbert-style Proof System

The Hilbert-style (or Frege) proof system for propositional logic comprises of 3 axiom schema and the modus ponens rule of inference

- **Axiom Schema 1:** $\varphi \rightarrow (\psi \rightarrow \varphi)$
- **Axiom Schema 2:** $(\varphi \rightarrow (\psi \rightarrow \rho)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \rho))$
- **Axiom Schema 3:** $((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$
- **Modus Ponens:** From φ and $\varphi \rightarrow \psi$, infer ψ

Question 1

Let Γ, Γ' be sets of propositional logic formulae and let φ be a formula. Show that if $\Gamma \vdash \varphi$ and if $\Gamma \subseteq \Gamma'$, then $\Gamma' \vdash \varphi$

Proof. $\Gamma \vdash \varphi \implies \exists$ a proof for $\phi, \langle \phi_1, \phi_2, \dots, \phi_n \rangle, \phi_n = \varphi$

Base case: $n = 1$

- Case 1: $\varphi \in \Gamma \implies \varphi \in \Gamma' \implies \Gamma' \vdash \varphi$
- Case 2: φ is from Axiom 1 or 2 or 3, $\vdash \varphi \implies \Gamma' \vdash \varphi$

Inductive case: Suppose $\forall i < n, \Gamma \vdash \phi_i \implies \Gamma' \vdash \phi_i$

- Case 1: φ is from axiom 1, 2, 3, trivial
- Case 2: φ is from Modus Ponens, $\exists i, j < n, \phi_i \equiv \phi_j \rightarrow \varphi, \Gamma \vdash \phi_i, \Gamma \vdash \phi_j$

$$\frac{\phi_j \rightarrow \varphi \quad \phi_j}{\varphi}$$

From inductive hypothesis, $\Gamma' \vdash \phi_i, \Gamma' \vdash \phi_j, \Gamma' \vdash \varphi$

□

Question 2

Theorem 0.1 (Deduction Theorem). *Let Γ be a set of propositional logic formulae and φ, ψ be propositional logic formulae. $\Gamma \vdash (\varphi \rightarrow \psi)$ iff $\Gamma \cup \{\varphi\} \vdash \psi$.*

Proof. ($\implies$)

1. $\Gamma \vdash \varphi \rightarrow \psi$ (Given)
2. $\Gamma \cup \{\varphi\} \vdash \varphi \rightarrow \psi$ (Monotonicity)
3. $\Gamma \cup \{\varphi\} \vdash \varphi$ (Assumption)
4. $\Gamma \cup \{\varphi\} \vdash \psi$ (MP 2, 3)

($\impliedby$)

Let $\pi = \psi_1, \dots, \psi_n$ be a proof of ψ from $\Gamma \cup \{\varphi\}$. Assume $\forall j < i, \Gamma \vdash \varphi \rightarrow \psi_j$.

Case 1: $\psi_i \in \Gamma$ or ψ_i is an Axiom

1. $\Gamma \vdash \psi_i$ (Assumption/Axiom)
2. $\Gamma \vdash \psi_i \rightarrow (\varphi \rightarrow \psi_i)$ (Ax 1)
3. $\Gamma \vdash \varphi \rightarrow \psi_i$ (MP 1, 2)

Case 2: $\psi_i = \varphi$

1. $\Gamma \vdash (\varphi \rightarrow ((\varphi \rightarrow \varphi) \rightarrow \varphi)) \rightarrow ((\varphi \rightarrow (\varphi \rightarrow \varphi)) \rightarrow (\varphi \rightarrow \varphi))$ (Ax 2)
2. $\Gamma \vdash \varphi \rightarrow ((\varphi \rightarrow \varphi) \rightarrow \varphi)$ (Ax 1)
3. $\Gamma \vdash (\varphi \rightarrow (\varphi \rightarrow \varphi)) \rightarrow (\varphi \rightarrow \varphi)$ (MP 1, 2)
4. $\Gamma \vdash \varphi \rightarrow (\varphi \rightarrow \varphi)$ (Ax 1)
5. $\Gamma \vdash (\varphi \rightarrow \varphi)$ (MP 3, 4)

Case 3: ψ_i via MP from ψ_j, ψ_k where $\psi_j = \psi_k \rightarrow \psi_i$

1. $\Gamma \vdash \varphi \rightarrow (\psi_k \rightarrow \psi_i)$ (IH)
2. $\Gamma \vdash (\varphi \rightarrow (\psi_k \rightarrow \psi_i)) \rightarrow ((\varphi \rightarrow \psi_k) \rightarrow (\varphi \rightarrow \psi_i))$ (Ax 2)
3. $\Gamma \vdash (\varphi \rightarrow \psi_k) \rightarrow (\varphi \rightarrow \psi_i)$ (MP 1, 2)
4. $\Gamma \vdash \varphi \rightarrow \psi_k$ (IH)
5. $\Gamma \vdash \varphi \rightarrow \psi_i$ (MP 3, 4)

□

Question 3

(a) $\vdash \varphi \rightarrow \varphi$

Done in Q2

(b) $\vdash (\psi \rightarrow \varphi) \rightarrow \varphi$, where $\psi = \varphi \rightarrow \perp$

Proof. $\equiv \{\psi \rightarrow \varphi\} \vdash \varphi$.

1. $(\varphi \rightarrow \perp) \rightarrow \varphi$ (Assumption)
2. $((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$ (Axiom 3)
3. $((\varphi \rightarrow \perp) \rightarrow \varphi) \rightarrow (((\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \perp)) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp))$ (Axiom 2)
4. $((\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \perp)) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)$ (MP 1, 3)
5. $(\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \perp)$ (Part a)
6. $(\varphi \rightarrow \perp) \rightarrow \perp$ (MP 4, 5)
7. φ (MP 2, 6)

(Wikipedia: Peirce's law).

□

(c) $\{\perp\} \vdash \varphi$

Proof. 1. $\perp$ (Assumption)

2. $\perp \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)$ (Axiom 1)
3. $(\varphi \rightarrow \perp) \rightarrow \perp$ (MP 1, 2)
4. $((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$ (Axiom 3)
5. φ (MP 3, 4)

□

(d) $\vdash \psi \rightarrow (\varphi \rightarrow \rho)$ **where** $\psi = \varphi \rightarrow \perp$

Proof. $\equiv \{\varphi \rightarrow \perp, \varphi\} \vdash \rho$.

1. $\varphi \rightarrow \perp$ (Assumption)
2. φ (Assumption)
3. $\perp$ (MP 1, 2)
4. $\perp \rightarrow ((\rho \rightarrow \perp) \rightarrow \perp)$ (Axiom 1)
5. $(\rho \rightarrow \perp) \rightarrow \perp$ (MP 3, 4)
6. $((\rho \rightarrow \perp) \rightarrow \perp) \rightarrow \rho$ (Axiom 3)
7. ρ (MP 5, 6)

□

Theorem 2 (Soundness of Hilbert-style Proof System)

Let Γ be a set of propositional logic formulae and φ be a propositional logic formula. If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$

Question 4

Prove Theorem 2. *Hint: Given a proof, inductively establish that every formula in the proof is a logical consequence of the assumptions.*

Proof. Let $\pi = \phi_1, \phi_2, \dots, \phi_n$ be a proof of φ from Γ , so $\phi_n = \varphi$

$$\Gamma \models \phi_i$$

$\forall 1 \leq i \leq n$.

1. If $\phi_i \in \Gamma \implies \Gamma \models \phi_i$.
2. If ϕ_i from an axiom, $\models \phi_i$
3. Otherwise, ϕ_i is by MP from ϕ_j and $\phi_k = \phi_j \rightarrow \phi_i$, where $j, k < i$. By IH

$$\Gamma \models \phi_j, \quad \Gamma \models \phi_j \rightarrow \phi_i.$$

$$\text{Let } v \models \Gamma, v \models \phi_j, v \models \phi_j \rightarrow \phi_i, v \models (\neg \phi_j \vee \phi_i), v \models \phi_i$$

Therefore $\Gamma \models \phi_n = \varphi$. Thus $\Gamma \vdash \varphi$ implies $\Gamma \models \varphi$. □

Theorem 3 (Completeness of Hilbert-style Proof System)

Let Γ be a set of propositional logic formulae and φ be a propositional logic formula. If $\Gamma \models \varphi$, then $\Gamma \vdash \varphi$

Theorem 4 (Weak Completeness of Hilbert-style Proof System)

Let φ be a propositional logic formula. If $\models \varphi$, then $\vdash \varphi$.

Definition 0.2 (Inconsistent set). Γ is inconsistent $\iff$

$$\exists \varphi, \Gamma \vdash \varphi, \Gamma \vdash (\varphi \rightarrow \perp)$$

Question 5

Prove that two notions of a set being inconsistent are identical. That is, prove that the following two statements are equivalent

1. There is a formula φ such that $\Gamma \vdash \varphi$ and $\Gamma \vdash (\varphi \rightarrow \perp)$
2. For every formula ρ , $\Gamma \vdash \rho$

Proof. (1 $\Rightarrow$ 2): If $\Gamma \vdash \varphi$ and $\Gamma \vdash \varphi \rightarrow \perp$, then by MP $\Gamma \vdash \perp$; since $\{\perp\} \vdash \rho$ for every formula ρ , by monotonicity we get $\Gamma \vdash \rho$ for every formula ρ .

(2 $\Rightarrow$ 1): for any formula φ , taking $\rho = \varphi$ and $\rho = \varphi \rightarrow \perp$ gives $\Gamma \vdash \varphi$ and $\Gamma \vdash \varphi \rightarrow \perp$. $\square$

Lemma 1

Let Γ be a set of formulae and φ be a formula. If $\Gamma \cup \{\varphi \rightarrow \perp\}$ is inconsistent, then $\Gamma \vdash \varphi$

Lemma 2

Let Γ be a set of formulae and φ be a formula. If $\Gamma \cup \{\varphi\}$ is inconsistent, then $\Gamma \vdash (\varphi \rightarrow \perp)$

Lemma 3

Let Γ be a set of formulae. If Γ is inconsistent, then there is a finite subset $\Gamma' \subseteq_{fin} \Gamma$ of Γ such that Γ' is also inconsistent

Question 6

- (a) Prove Lemma 1

Proof. From deduction thm, $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow \psi$ and $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow (\psi \rightarrow \perp)$.

1. $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow (\psi \rightarrow \perp)$
2. $\Gamma \vdash ((\varphi \rightarrow \perp) \rightarrow (\psi \rightarrow \perp)) \rightarrow (((\varphi \rightarrow \perp) \rightarrow \psi) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp))$ (Ax 2)
3. $\Gamma \vdash ((\varphi \rightarrow \perp) \rightarrow \psi) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)$ (MP 1, 2)
4. $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow \psi$
5. $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow \perp$ (MP 3, 4)
6. $\Gamma \vdash ((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$ (Ax 3)
7. $\Gamma \vdash \varphi$ (MP 5, 6)

$\square$

- (b) Prove Lemma 2

Proof. By definition, $\Gamma \cup \{\varphi\} \vdash \psi$ and $\Gamma \cup \{\varphi\} \vdash (\psi \rightarrow \perp)$ for some ψ . By the Deduction:

1. $\Gamma \vdash \varphi \rightarrow \psi$
2. $\Gamma \vdash \varphi \rightarrow (\psi \rightarrow \perp)$
3. $\Gamma \vdash (\varphi \rightarrow (\psi \rightarrow \perp)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \perp))$ (Ax 2)
4. $\Gamma \vdash (\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \perp)$ (MP 2, 3)
5. $\Gamma \vdash \varphi \rightarrow \perp$ (MP 1, 4)

$\square$

- (c) Let Γ be a set of formulae and φ be a formula. Show that, if Γ is consistent, then at least one of $\Gamma \cup \{\varphi\}$ or $\Gamma \cup \{\varphi \rightarrow \perp\}$ is consistent

Proof. Suppos otherwise, Γ is consistent, $\Gamma \cup \{\varphi\}$ and $\Gamma \cup \{\varphi \rightarrow \perp\}$ are inconsistent. Since $\Gamma \cup \{\varphi\}$ is inconsistent, by Lemma 2

$$\Gamma \vdash \varphi \rightarrow \perp .$$

Since $\Gamma \cup \{\varphi \rightarrow \perp\}$ is inconsistent, by Lemma 1 we have

$$\Gamma \vdash \varphi .$$

$$\Gamma \vdash \perp .$$

By explosion, pick an arbitrary formula ψ , $\Gamma \vdash \perp \vdash \psi$, $\Gamma \vdash \perp \vdash \psi \rightarrow \perp$, Γ is inconsistent, contradiction. $\square$

- (d) Let Γ, Γ' be sets of formulae such that $\Gamma \subseteq \Gamma'$. If Γ' is consistent, then so is Γ

Proof. Suppose othrewise, Γ is inconsistent,

$$\Gamma \vdash \perp .$$

$\Gamma \subseteq \Gamma'$, by monotonicity of derivability we have

$$\Gamma' \vdash \perp .$$

Contradiction. $\square$

- (e)

Proof. Γ is inconsistent, $\exists \varphi, \Gamma \vdash \varphi, \Gamma \vdash (\varphi \rightarrow \perp)$ there exists a finite proof for both φ and $\varphi \rightarrow \perp$. Consider

- $\pi_1 = \langle \phi_1, \phi_2, \dots, \phi_n \rangle, \phi_n = \varphi$
- $\pi_2 = \langle \psi_1, \psi_2, \dots, \psi_m \rangle, \psi_m = \varphi \rightarrow \perp$

We construct Γ' as such: $\forall i < n, j < m$, if $\phi_i \in \Gamma, \phi_i \in \Gamma'$; if $\psi_j \in \Gamma, \psi_j \in \Gamma'$ Therefore $\forall \eta \in \Gamma', \eta \in \Gamma, \Gamma' \subseteq \Gamma$; $|\Gamma'| \leq n + m$, Γ' is a finite subset of Γ . Since π_1 is a proof for ϕ , $\forall \phi \in \pi_1, \phi \in \Gamma \implies \phi \in \Gamma'$; We can easily show that same proof can be constructed from Γ' by induction. Same case for $\Gamma' \vdash \varphi \rightarrow \perp$. Therefore Γ' is inconsistent. $\square$

Definition 0.3. • $\Delta_0 = \{\varphi \rightarrow \perp\}$

- $\Delta_{i+1} = \begin{cases} \Delta_i \cup \{\varphi_i\} & \text{if } \Delta_i \cup \{\varphi_i\} \text{ is consistent} \\ \Delta_i \cup \{\varphi_i \rightarrow \perp\} & \text{otherwise} \end{cases}$

$$\Delta = \bigcup_{i \geq 0} \Delta_i$$

Lemma 0.4. Γ is consistent $\implies \Gamma \cup \{\varphi\}$ or $\Gamma \cup \{\varphi \rightarrow \perp\}$ is consistent.

Question 7

- (a) Prove, using induction, that Δ_i is consistent, for every i

Proof. **Base case:** Δ_0

Suppose otherwise, $\Delta_0 = \{\varphi \rightarrow \perp\}$ is inconsistent. Then by Lemma 1,

$$\vdash \varphi .$$

Contradiction.

Inductive case: Suppose $\forall j \leq i, \Delta_j$ is consistent

By lemma above, Δ_i is consistent $\implies \Delta_i \cup \{\varphi_i\}$ is consistent or $\Delta_i \cup \{\varphi_i \rightarrow \perp\}$ is consistent $\implies \Delta_{i+1}$ is consistent $\square$

- (b) Prove that Δ is consistent

Proof. Suppose otherwise, Δ is inconsistent, by Lemma 3, $\exists \Delta' \subseteq_{fin} \Delta, \Delta'$ is inconsistent. Since Δ' is finite, let the highest ranked formula in Δ' be $n, \forall \varphi \in \Delta', \varphi \in \Delta_n$. From b), Δ_n is consistent, by 6d), Δ' is consistent, contradiction. $\square$

- (c) Prove that Δ is negation-complete, i.e., for every formula ψ , we have $(\psi \rightarrow \perp) \in \Delta$ iff $(\psi \notin \Delta)$

Proof. $\implies$

Let $\psi = \varphi_{n+1}, \Delta_n$ is consistent. $\Delta_{n+1} = \Delta_n \cup \{\psi\}$ if $\Delta_n \cup \{\psi\}$ is consistent. Suppose $\psi \rightarrow \perp \in \Delta$, $\Delta \vdash \psi, \Delta \vdash \psi \rightarrow \perp \implies \Delta$ is inconsistent, contradiction. Therefore $\psi \in \Delta \implies \psi \rightarrow \perp \notin \Delta$

$\Leftarrow$

Suppose otherwise, $\psi \rightarrow \perp \in \Delta, \psi \in \Delta, \Delta \vdash \psi, \Delta \vdash \psi \rightarrow \perp \implies \Delta$ is inconsistent, contradiction. $\square$

- (d) Use the fact that Δ is negation-complete to show that for every formula ψ , $\psi \in \Delta$ iff $\Delta \vdash \psi$

Proof. $\implies$

trivial by definition

$\Leftarrow$

Consider $\Delta \vdash \psi$. Suppose $\psi \notin \Delta, \psi \rightarrow \perp \in \Delta, \Delta \vdash \psi \rightarrow \perp$. Since $\Delta \vdash \psi$, by MP, $\Delta \vdash \perp$, Δ is inconsistent, contradiction. Therefore $\psi \in \Delta$. $\square$

Question 8

- (a) Show that for any formula ψ , $v \models \psi$ iff $\psi \in \Delta$

Proof. Base case:

1. $\psi \in P$: by definition, $v(p) = \text{true} \iff p \in \Delta$
2. $\psi = \perp$: always $v(\perp) = \text{false}, \perp \notin \Delta \implies v \models \perp \iff \perp \in \Delta$ vacuously

Inductive case: $\psi = a \rightarrow b$

Suppose

- $v \models a \iff a \in \Delta$
- $v \models b \iff b \in \Delta$

Case $\implies$

Suppose $v \models a \rightarrow b$, $a \rightarrow b \notin \Delta$, $(a \rightarrow b) \rightarrow \perp \in \Delta$, $\Delta \vdash (a \rightarrow b) \rightarrow \perp$.

Now we show that $b \notin \Delta$

- Suppose otherwise, $b \in \Delta$, $\Delta \vdash b$, by ax 1, $\Delta \vdash b \rightarrow (a \rightarrow b) \implies \Delta \vdash a \rightarrow b$ by MP, contradiction.

Now we show that $a \in \Delta$

- Suppose otherwise, $a \notin \Delta \implies a \rightarrow \perp \in \Delta$, from

$$\vdash (\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \rho)$$

$\Delta \vdash (a \rightarrow \perp) \rightarrow (a \rightarrow b)$. By MP, $\Delta \vdash a \rightarrow b$, contradiction.

Now with $b \notin \Delta$ and $a \in \Delta$, by IH, $v \models a$ and $v \not\models b \implies v \not\models a \rightarrow b$, contradiction.

Case $\Leftarrow$

Suppose otherwise, $a \rightarrow b \in \Delta$, $v \models a$ and $v \not\models b$. By IH, $a \in \Delta$ and $b \notin \Delta$, $b \rightarrow \perp \in \Delta \implies \Delta \vdash a, \Delta \vdash a \rightarrow b$, by MP, $\Delta \vdash b \implies b \in \Delta$, contradiction. $\square$

- (b) Show that $v \models (\varphi \rightarrow \perp)$ and thus $\not\models \varphi$

Proof. $\{\varphi \rightarrow \perp\} \in \Delta \implies v \models (\varphi \rightarrow \perp) \implies v \not\models \varphi \implies \not\models \varphi$. $\square$

Question 9

Prove Theorem 3 using Theorem 4, the Compactness theorem and the Deduction theorem

Proof. $\Gamma \models \varphi \implies \Gamma \cup \{\varphi \rightarrow \perp\}$ is unsat, so by Compactness $\exists \Gamma_0 = \{\psi_1, \dots, \psi_n\} \subseteq_{\text{fin}} \Gamma$, such that $\Gamma_0 \cup \{\varphi \rightarrow \perp\}$ is unsat,

$$\{\psi_1, \dots, \psi_n, (\varphi \rightarrow \perp)\} \text{ is unsat } \implies (\psi_1 \wedge \dots \wedge \psi_n \wedge (\varphi \rightarrow \perp)) \models \perp$$

$$\models \psi_1 \rightarrow (\psi_2 \rightarrow \dots (\psi_n \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)) \dots)$$

([https://en.wikipedia.org/wiki/Exportation_\(logic\)](https://en.wikipedia.org/wiki/Exportation_(logic))) by Theorem 4

$$\emptyset \vdash \psi_1 \rightarrow (\psi_2 \rightarrow \dots (\psi_n \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)) \dots)$$

Apply deduction thm

$$\{\psi_1\} \vdash (\psi_2 \rightarrow \dots (\psi_n \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)) \dots)$$

until we get

$$\Gamma_0 \vdash (\varphi \rightarrow \perp) \rightarrow \perp.$$

Using ax 3:

$$\vdash ((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi,$$

By MP

$$\Gamma_0 \vdash \varphi.$$

Since $\Gamma_0 \subseteq \Gamma$, by monotonicity,

$$\Gamma \vdash \varphi.$$

$\square$

Theorem 5 (Soundness of Resolution Proof System)

Let Γ be a set of clauses (over propositions P). If there is a resolution proof of Γ , then Γ is unsatisfiable.

Question 10

Prove Theorem 5, *Hint: Consider a resolution proof $\pi = C_1, C_2, \dots, C_n$ of Γ . Use induction to establish that for every $1 \leq i \leq n$ if a valuation v is such that $v \models \Gamma$, then $v \models C_i$.*

Proof. Let $\pi = C_1, C_2, \dots, C_n$ be a resolution proof of Γ , where $C_n = \emptyset$.

We prove by induction on i that for every $1 \leq i \leq n$, $v \models \Gamma, v \models C_i$.

$i \in \{1, \dots, n\}$ and suppose $v \models \Gamma$.

1. $C_i \in \Gamma, v \models C_i$.
2. C_i is a resolvent of two earlier clauses C_j and C_k wrt some proposition p , where

$$C_j = C \uplus \{p\}, \quad C_k = D \uplus \{\neg p\},$$

and $C_i = C \cup D$. By the IH, $j, k < i, v \models C_j$ and $v \models C_k$. We show that $v \models C_i$. Suppose, $v \not\models C_i = C \cup D$. Then v satisfies neither any literal in C nor any literal in D . Since $v \models C_j = C \uplus \{p\}$ and no literal in C is satisfied by v , it must be that $v \models p$, i.e. $v(p) = \text{true}$. Since $v \models C_k = D \uplus \{\neg p\}$ and no literal in D is satisfied by v , it must be that $v \models \neg p$, i.e. $v(p) = \text{false}$, Contradiction

□

Theorem 6 (Completeness of Resolution Proof System)

Let Γ be a set of clauses (over propositions P). If Γ is unsatisfiable, then there is a resolution proof of Γ

Question 11

Let Γ be an unsatisfiable set of clauses. Show that, if Γ is unsatisfiable, then there is a finite set $\Gamma_0 \subseteq_{fin} \Gamma$ such that Γ_0 is unsatisfiable.

Proof. Suppose Γ is unsat, by Compactness, $\exists \Gamma' \subseteq_{fin} \Gamma, \Gamma'$ is unsat (?)

□

Question 12

Let Γ_1, Γ_2 be sets of clauses such that $\Gamma_1 \subseteq \Gamma_2$. Show that, if there is a resolution proof of Γ_1 , then there is also a resolution proof of Γ_2

Proof. Let $\pi = C_1, C_2, \dots, C_n$ be a resolution proof of Γ_1

$\forall 1 \leq i \leq n, \pi$ is a resolution proof of Γ_1 , either:

1. $C_i \in \Gamma_1$, or
2. $\exists j, k < i, C_i$ is a resolvent of C_j and C_k

If $C_i \in \Gamma_1, C_i \in \Gamma_2$. Otherwise, this fact is independent of the choice of the initial clause set, so the same derivation is valid in a proof from Γ_2 .

□

Question 13

Next, we constructed a sequence of sets as follows. Here, we assume that the set of propositions occurring in Γ_0 can be enumerated as $p_1, p_2, \dots, p_k$.

$$\Delta_0 = \Gamma_0$$

$$\Delta_{i+1} = \Delta_i^\emptyset \cup \{C \cup D \mid C \uplus \{p_{i+1}\} \in \Delta_i^+, D \uplus \{\neg p_{i+1}\} \in \Delta_i^-\}$$

where

- $\Delta_i^\emptyset$ is the set of clauses in Δ_i that contain neither the literal p_{i+1} nor $\neg p_{i+1}$,
- Δ_i^+ is the set of clauses in Δ_i that contain the literal p_{i+1} but not $\neg p_{i+1}$,
- Δ_i^- is the set of clauses in Δ_i that contain the literal $\neg p_{i+1}$ but not p_{i+1} .

Inductively show that Δ_i is unsat for every $0 \leq i \leq k$. Also show that $\Delta_i \neq \emptyset$ for every $0 \leq i \leq k$.

Proof. **Base case:**

$i = 0$. By construction, $\Delta_0 = \Gamma_0$. From 11, Γ_0 is a finite unsat subset of Γ

Inductive step:

Suppose Δ_i is unsat and $\Delta_i \neq \emptyset$. We show Δ_{i+1} is unsat and nonempty. Suppose that $\exists v, v \models \Delta_{i+1}$, consider $v(p_{i+1})$

- If $v(p_{i+1}) = \text{true}$, then every clause in Δ_i^+ is sat by v due to p_{i+1} , and $\Delta_i^\emptyset \subseteq \Delta_{i+1}$ thus sat by v .

For every resolvent $C \cup D \in \Delta_{i+1}$ with

$$C \uplus \{p_{i+1}\} \in \Delta_i^+, \quad D \uplus \{\neg p_{i+1}\} \in \Delta_i^-,$$

$v \models C \cup D$. Since $v(p_{i+1}) = \text{true}$, the clause $D \uplus \{\neg p_{i+1}\}$ must be satisfied by some literal in D , hence $v \models D$. Thus v satisfies every clause in Δ_i^- . Contradiction

- case $v(p_{i+1}) = \text{false}$: similar

□

Question 14

Inductively show that Δ_i does not contain the literals $\{p_1, \neg p_1, \dots, p_i, \neg p_i\}$ for every $1 \leq i \leq k$

Proof. We prove by induction on i that Δ_i does not contain any of the literals

$$\{p_1, \neg p_1, \dots, p_i, \neg p_i\}$$

for every $1 \leq i \leq k$.

Base case:

$i = 1$.

By definition,

$$\Delta_1 = \Delta_0^\emptyset \cup \{C \cup D \mid C \uplus \{p_1\} \in \Delta_0^+, D \uplus \{\neg p_1\} \in \Delta_0^-\}.$$

Every clause in $\Delta_0^\emptyset$ contains neither p_1 nor $\neg p_1$, and every resolvent $C \cup D$ is obtained by removing p_1 and $\neg p_1$. Hence no clause in Δ_1 contains p_1 or $\neg p_1$.

Inductive step:

□

Question 15

Finish the proof of Theorem 6 by explicitly constructing a resolution proof for Γ and establishing its correctness

Proof. Let Γ be unsat. By 11, $\exists \Gamma_0 \subseteq_{fin} \Gamma$ such that Γ_0 is unsat. Let $p_1, \dots, p_k$ be all propositions occurred in Γ_0 , and $\Delta_0, \Delta_1, \dots, \Delta_k$ as in the construction. Δ_k is unsat, and by 14, Δ_k does not contain $p_1, \neg p_1, \dots, p_k, \neg p_k$. therefore every clause in Δ_k contains no literal, thus every clause in Δ_k is $\emptyset$. Since $\Delta_k \neq \emptyset$, $\emptyset \in \Delta_k$

Base case:

$i = 0$, $\Delta_0 = \Gamma_0$

Inductive step:

Suppose every clause in Δ_i has a finite resolution proof from Γ_0 . Let $E \in \Delta_{i+1}$. By definition ,

1. $E \in \Delta_i^\emptyset$, hence $E \in \Delta_i$, so by IH E has a finite resolution proof from Γ_0 ; or
2. $E = C \cup D$, where

$$C \uplus \{p_{i+1}\} \in \Delta_i^+ \quad \text{and} \quad D \uplus \{\neg p_{i+1}\} \in \Delta_i^-.$$

Then E is the resolvent of these 2

$$C \uplus \{p_{i+1}\} \quad \text{and} \quad D \uplus \{\neg p_{i+1}\}.$$

By IH, both clauses have a finite resolution proof from Γ_0 , thus the resolution proof for E is also finite

$\Gamma_0 \subseteq \Gamma$, by 12 the same proof is also a resolution proof for Γ .

Therefore Γ is unsat, then there is a resolution proof for Γ . □

We will prove the contrapositive of compactness theorem for the next 2 qn.

Question 16

Assume that the Hilbert-style Proof System is sound and complete for propositional logic, Use these facts to establish that the Compactness Theorem holds true for propositional logic

Proof. Γ is unsat $\implies \Gamma \models \perp$, so by completeness $\Gamma \vdash \perp$ since proofs are finite, only finitely many assumptions from Γ appear in the proof, yielding a finite $\Gamma_0 \subseteq_{\text{fin}} \Gamma$ with $\Gamma_0 \vdash \perp$, and by soundness $\Gamma_0 \models \perp$, hence Γ_0 is unsat □

Question 17

Assume that the Resolution Proof System is sound and complete for propositional logic. Use these facts to establish that the Compactness Theorem holds true for propositional logic

Proof. If Γ is an unsatisfiable set of clauses, by completeness there is a finite resolution proof for Γ , and since this proof uses only finitely many clauses from Γ , there is a finite subset $\Gamma_0 \subseteq_{\text{fin}} \Gamma$ from which the same proof is a resolution proof. By soundness, Γ_0 is unsat □